

4.3

Enterprise Data Warehouse

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OBJECTIVE

This exercise shows how to create an enterprise data warehouse. It illustrates how to create advanced InfoObjects such as an aDSO needed for loading transactional data and a composite provider needed for reporting.

ACTIVITIES

- Learn how to create an advanced DataStore Object and how to load it with data.
- Learn how to create the virtualization layer of the data warehouse.

SOFTWARE PREREQUISITES

- Eclipse with SAP BW Modeling Tools
- SAP GUI

DATA SET

Global Bike (GB) data set

SYSTEM ACCESS

SAP BW/4HANA 2.0 from your UCC

Scenario

Jan has already loaded the transaction data from the source system into the data acquisition layer (exercise 4.2), modelled InfoObjects as elementary information objects (exercise 3.2) and loaded the master data (exercise 4.1). In this exercise Jan will model the data warehouse and data mart layer and load the transaction data from the data acquisition layer.

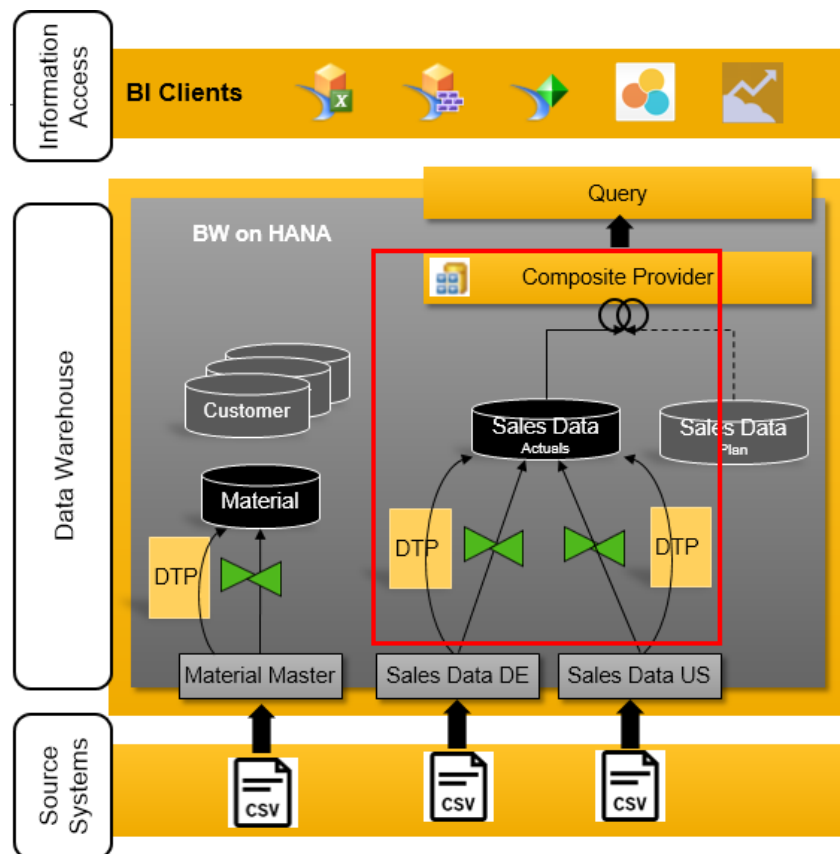


Figure 1

In a previous exercise (2) the sales transaction data were stored in the data acquisition layer. In the next steps it will be loaded into the data warehouse. First you need an advanced DataStore Object (aDSO) based on the sales transaction data DataSource. Then you will load the sales transaction data from Global Bike Germany and US in the data warehouse. Finally you will create a Composite Provider as a virtualization layer for analytics.

The graphic below displays the complete process (8 tasks).

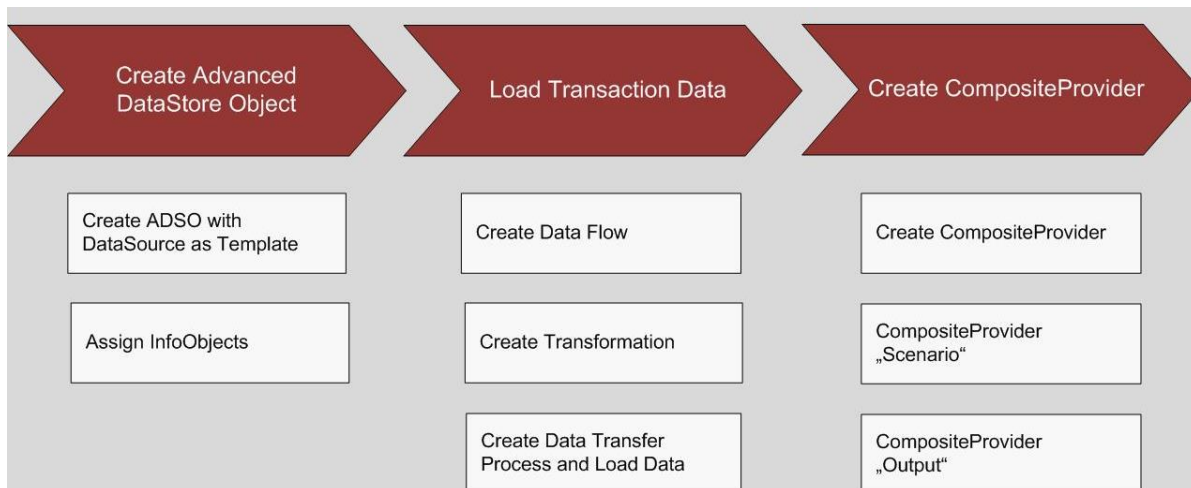


Figure 2

1. Create aDSO with DataSource as a Template

Jan uses Eclipse to create an *Advanced DataStore Object (aDSO)* for sales transaction data of Global Bike Germany and US.

1. Start Eclipse and open your BW project: **Project Explorer** → **your BW-System** → **BW Repository**.

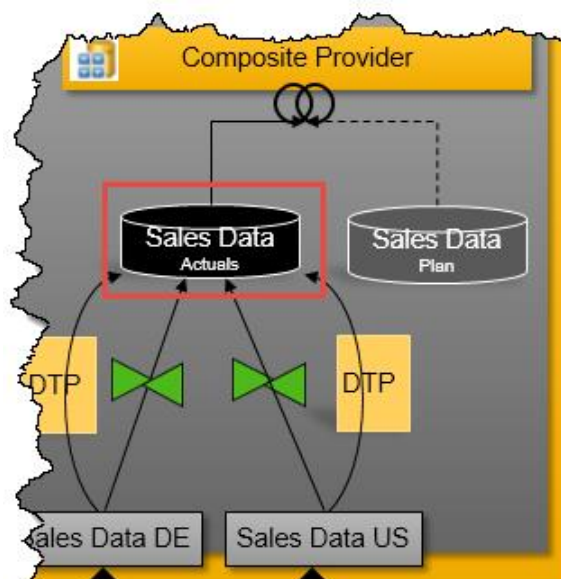


Figure 3

2. Log on and navigate to the InfoArea assigned by your instructor, which has the technical name **XXX###** and description **Student ###**.
3. Right-click and choose **New** → **DataStore Object (advanced)**.

Note: When working with the exercise, please replace in all names XXX with the Id of your course, and ### with your user number (three digits) provided by your instructor.

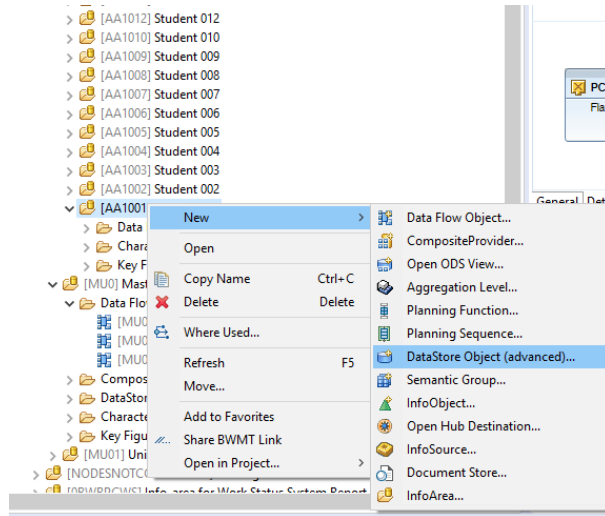


Figure 4

4. On the following screen, name **XXXDM1###** and description **XXX### Sales Data aDSO**. Use one of your DataSources from exercise 4.2 Data Acquisition as a template (e.g. **XXXDSDE###**).
5. All fields of your DataSource are copied as fields to DataStore Object (includes type information and other metadata).

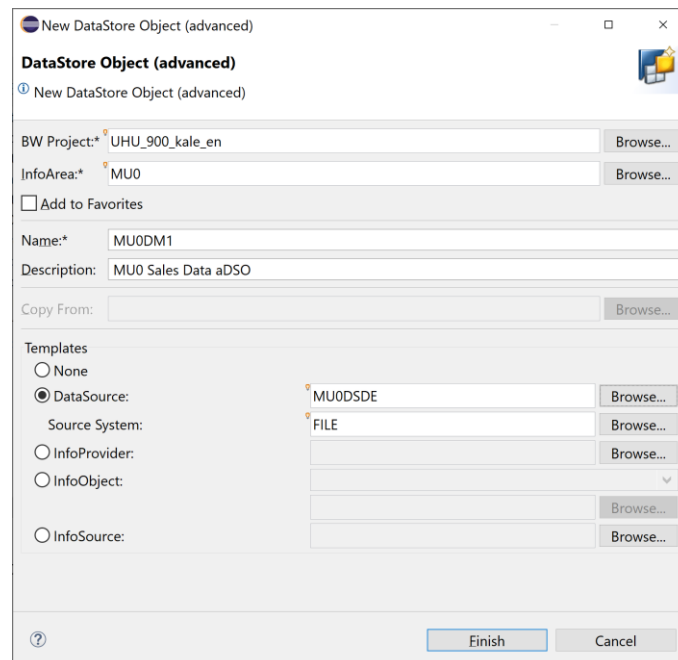


Figure 5

6. Finish.

- On **General** tab, check **Data Mart DataStore Object** option. This sets some general properties in the aDSO, like activation of data after updates or key fields of the aDSO.

General: MU0DM1

DataStore Object (advanced)	
Technical Name:	MU0DM1
Description:	MU0 Sales Data aDSO
☐ External SAP HANA View ☐ Log Read Access Output	
Modeling Properties	Data Tiering Properties
☐ Standard DataStore Object ☐ Write Change Log ☐ Snapshot Support ☐ Unique Data Records ☐ Staging DataStore Object ☐ Inbound Queue Only ☐ Compress Data ☐ Reporting-Enabled ☒ Data Mart DataStore Object ☐ Direct Update DataStore Object	Temperature Schema / Tiers ☒ Hot (SAP HANA Standard Nodes) ☐ Warm (SAP HANA Extension Nodes) ☐ Cold (External Cold Store) Connection: Browse... Temperature Maintenance ☒ On Object Level ☐ On Partition Level

Figure 6

- Save your aDSO.

2. Assign various InfoObjects to your Advanced DataStore Object.

Jan uses Eclipse to assign various InfoObjects to the aDSO.

- Start **Eclipse** and open your BW project: **Project Explorer** → **your BW System** → **BW Repository**.
- If you closed Eclipse, log on again and navigate to the InfoArea assigned by your instructor, which has the technical name **XXX###** and description **Student ###**.
- Open your aDSO from the previous step (technical name: **XXXDM1###**).
- On **Details** tab you see a list of all fields that have been copied from the DataSource template. On this tab, we will model the following:
 - Use InfoObjects instead of fields:** this step for creating valuable information out of data (“data to information”), as the data source is just the inbound representation of an external source of data, whereas InfoObjects represent company wide, integrated, and harmonized information objects.
 - Add InfoObjects** that enrich our aDSO with information that were not a part of the original.

5. We will group the fields/InfoObjects into groups for a better overview. The groups correspond to dimensions in a multidimensional model.
6. First, create 5 groups: **ORDER**, **CUSTOMER**, **PRODUCT**, **TIME**, and **MEASURES**.

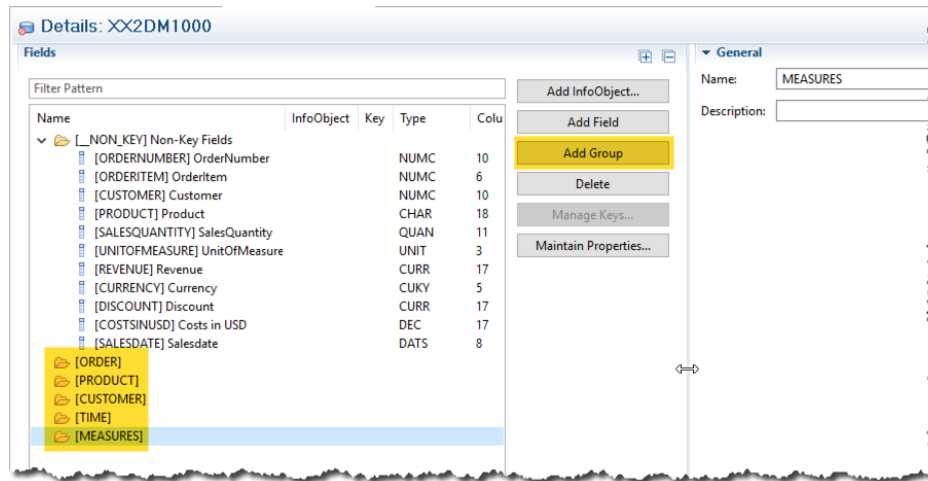


Figure 7

7. **Save.**
8. Then move the fields **ORDERNUMBER** & **ORDERITEM** into **ORDER**. For these fields, there are no corresponding InfoObjects in the system. *Fields have a different icon than InfoObjects. Pay attention to the difference.*
9. In the following steps, you will change fields into corresponding InfoObjects.

Hint: Make use of the naming convention to find out the technical names of the corresponding InfoObjects.

- All InfoObjects delivered by SAP have a name starting with **O**....
 - All InfoObjects delivered by UCC have a name starting with **MUO**...
 - All InfoObjects, you created have a name starting with your **XXX**...
10. Click on the **Customer** field. Then click **Change Type** and then browse for **MUoCUST**. This is how it should change from Field to InfoObject. Note its icon.
 11. Then use the following table to change fields into InfoObjects.

Details: AA1DM1001 Revised Version

Fields

Filter Pattern

Name	Key	Type	Length/ Pre
[NON_KEY] Non-Key Fields			
[MUOCUST] Customer		CHAR	10
[COUNTRY] Country		CHAR	2
[PRODUCT] Product		CHAR	18
[SALESQUANTITY] SalesQuantity		QUAN	15:0
[UNITOFMEASURE] UnitOfMeasure		UNIT	3
[REVENUE] Revenue		CURR	17:2
[CURRENCY] Currency		CUKY	5
[DISCOUNT] Discount		CURR	17:2
[COSTSUSD] CostsUSD		DEC	17:2
[SALESDATE] SalesDate		DATS	8

[ORDER]
 [PRODUCT]
 [CUSTOMER]
 [TIME]
 [MEASURES]

Add InfoObjects...
 Add Field
 Add Group
 Remove
 Manage Keys...
 Maintain Properties...

General

Type: InfoObject Change Type...

Name: MUOCUST Browse...

Description: Customer

Data Type

Data Type: CHAR Length: 10

Properties

Aggregation: NONE

Unit/Currency Field:

Fixed Unit/Currency:

Master Data Check: Master Data Check During Load/

Conversion Routine: ALPHA Browse...

☐ Use Navigation Attributes in Extraction
☐ Use Characteristic as Key Figure

Criteria (0)

Compoundings (0)

Figure 8

Group	Field	Change Field to the following InfoObject
CUSTOMER	Customer	Customer (delivered by UCC)
PRODUCT	Product	Material (created by you)
TIME	Salesdate	Calendar Day (delivered by SAP)
MEASURES	UnitofMeasure	OUNIT
MEASURES	SalesQuantity	Sales Quantity (created by you)
MEASURES	Revenue	Revenue (created by you)
MEASURES	Discount	Discount (delivered by UCC)
MEASURES	CostsInUSD	Cost of Goods manufactured USD (delivered by UCC)

12. Now **move** the InfoObjects into the corresponding group according to the following table.

GROUP	Name	Technical Name
Non-Key Fields	Unit of Measure	UNITOFMEASURE
	Currency	CURRENCY
ORDER	OrderItem	ORDERITEM
	OrderNumber	ORDERNUMBER
PRODUCT	Material	XXXMA###
CUSTOMER	Customer	MUOCUST
TIME	Calendar Day	OCALDAY

MEASURE	CostUSD	MU0COGM
	Discount	MU0DISCNT
	Revenue	XXXREV###
	Currency Key	OCURRENCY
	Sales Quantity	XXXQTY###
	Unit of Measure	OUNIT

13. Move the InfoObject **Currency Key** to **CUSTOMER**.

14. Move the InfoObject **Unit of Measure** to **PRODUCT**.

15. This is how your aDSO should look:

Name	Key	Type
▼ [ORDER]		
[ORDERNUMBER] OrderNumber		NUMC
[ORDERITEM] OrderItem		NUMC
▼ [PRODUCT]		
[OUNIT] Unit of Measure		UNIT
[AB1MA191] AB1191 Material		CHAR
▼ [CUSTOMER]		
[OCURRENCY] Currency Key		CUKY
[MU0CUST] Customer		CHAR
▼ [TIME]		
[OCALDAY] Calendar Day		DATS
▼ [MEASURES]		
[MU0COGM] Cost of Goods Manufactured USD		CURR
[MU0DISCNT] Discount		CURR
[AB1REV191] AB1191 Revenue		CURR
[AB1QTY191] AB1191 Sales Quantity		QUAN

Figure 9

16. After this, remove all remaining Non-Key Fields (**Currency**, **UnitOfMeasure**) and then the folder **Non-Key Fields**.

17. Add more **InfoObjects** as the following characteristics and measures to your aDSO:

Group	InfoObject
MEASURES	Net Sales (delivered by UCC)
CUSTOMER	Sales Organization (delivered by UCC)
TIME	Calendar year (delivered by SAP)
TIME	Calendar month (delivered by SAP)
TIME	Calendar year/month (delivered by SAP)

Additional Information of handling of currency & unit: When you identify Key Figures like revenue or sales quantity, corresponding InfoObjects for currency (oCURRENCY) or unit of measurement (oBASEUOM) are added automatically by the system.

18. Move **oCURRENCY** to group **CUSTOMER**, and **oUNIT** to group **PRODUCT**, as they semantically belong to these entities.
19. Your aDSO should look like this (replace with appropriate InfoObjects that you created in previous exercise). Pay attention to the icons and technical names. *Correct any errors now otherwise your ETL process will fail in later steps!*

Name	Key	Type
▼ [ORDER]		
[ORDERITEM] OrderItem		NUMC
[ORDERNUMBER] OrderNumber		NUMC
▼ [PRODUCT]		
[0UNIT] Unit of Measure		UNIT
[AA1MA001] AA1001 Material		CHAR
▼ [CUSTOMER]		
[MU0SALORG] Sales Organization		CHAR
[0CURRENCY] Currency Key		CUKY
[MU0CUST] Customer		CHAR
▼ [TIME]		
[0CALYEAR] Calendar Year		NUMC
[0CALMONTH2] Calendar month		NUMC
[0CALMONTH] Calendar Year/Month		NUMC
[0CALDAY] Calendar Day		DATS
▼ [MEASURES]		
[MU0NETSAL] Net Sales		CURR
[MU0COGM] Cost of Goods Manufactured USD		CURR
[MU0DISCNT] Discount		CURR
[AA1REV001] AA1001 Revenue		CURR
[AA1QYT001] AA1001 Sales Quantity		QUAN

Figure 10

20. Finally, **save** and **activate** your aDSO. If everything is correct, you get the following message:

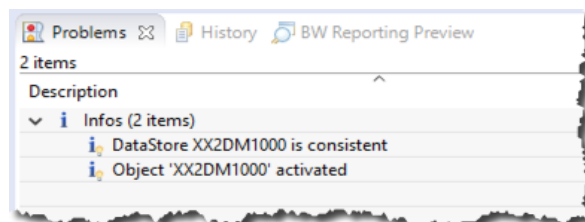


Figure 11

We have now modelled our main persistency, the semantically enriched data mart for worldwide sales data of Global Bike. As a new paradigm in-memory computing with SAP

BW/4HANA, all data are persisted on a detailed level and all aggregation is done on the fly. There is now multidimensional storage the advanced DataStore Object.

21. **Close** the aDSO.

Question 1: What is the difference between a field and an InfoObject?

Question 2: What is the advantage of using InfoObjects instead of fields in an aDSO?

3. Create a data flow object.

Create a data flow object to model the loading of sales transaction data for Global Bike Germany and US. Add the *Sales Transaction Data DataSource* from Global Bike Germany and US. Then, add the aDSO to the data flow.

1. Open **Eclipse**
2. Navigate to your InfoArea folder: **XX** → **XXX** → **XXX###** (name: **Student ###**).
3. Choose **New** → **Data Flow Object** in the context menu (right-click).
4. For the technical name: **XXXDFSD###** and as description **XXX### Data Flow Sales Data**.
5. **Finish**.
6. Drag and drop the **DataSource** icon (from the right panel **BW Object Types**) into your Data Flow.
7. To insert your DataSource choose **Add Object** in the context menu of the object that you just added.
8. Search and select your German Sales Data DataSource with the technical name **XXXDSDE###**.
9. Follow the steps above to also add your US Sales data datasource with the technical name **XXXDSUS###**.
10. Finally, add your aDSO **XXXDM1###** from previous step to the data flow.
11. After finishing this, your Data Flow should look like the picture below.

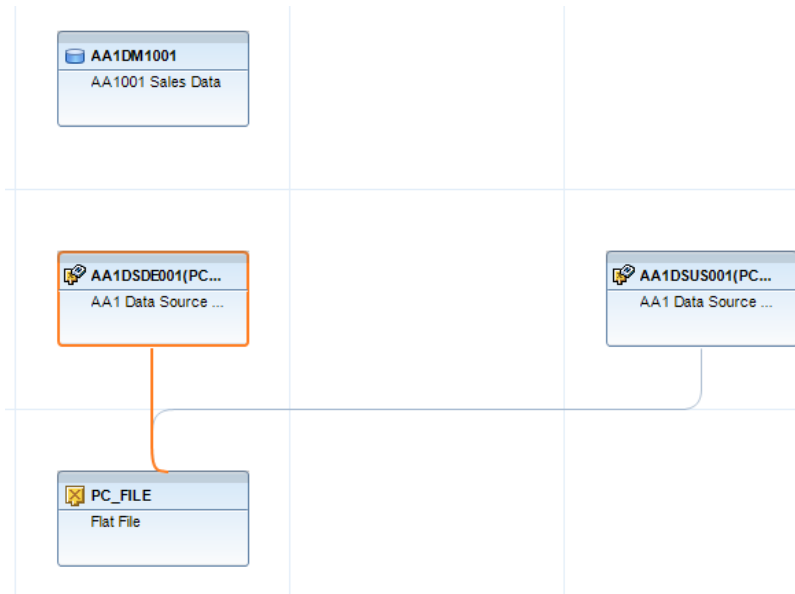


Figure 12

12. **Save** and **activate** the data flow.

Question 3: Insert a screenshot of your Data Flow and label each box with the type of object it is (not the name).

4. Create and define rules for a transformation.

Now you will define the *transformation* rules between the data acquisition layer (sales transaction Data Source from Global Bike Germany and US) and the data mart layer (advanced DataStore Object).

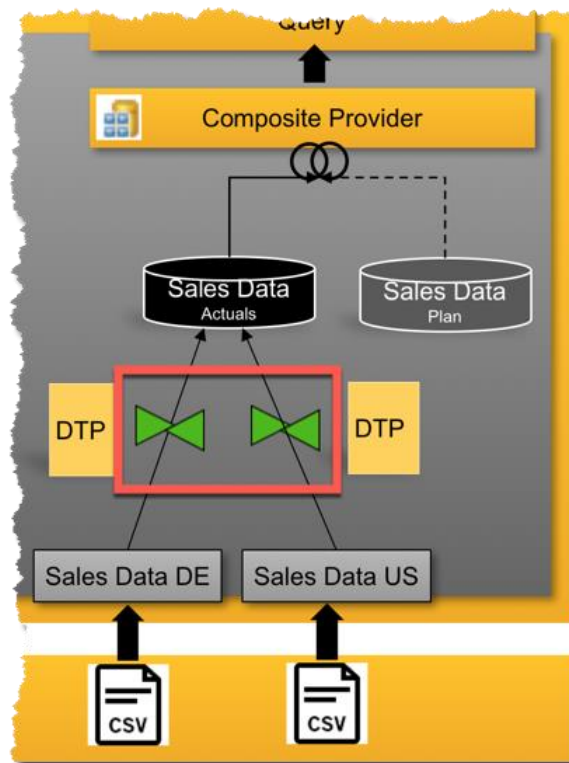


Figure 13

1. Open **Eclipse**
2. Navigate to your InfoArea folder: **XX** → **XXX** → **XXX###** (name: **Student ###**).
3. Open your Data Flow from previous step with technical name **XXXDFSD###**.
4. **Connect** the left DataSource (**Sales Data DE ###**) with the advanced DataStore Object (**XXX### Sales Data**) by drawing a line between the two objects using the arrow.

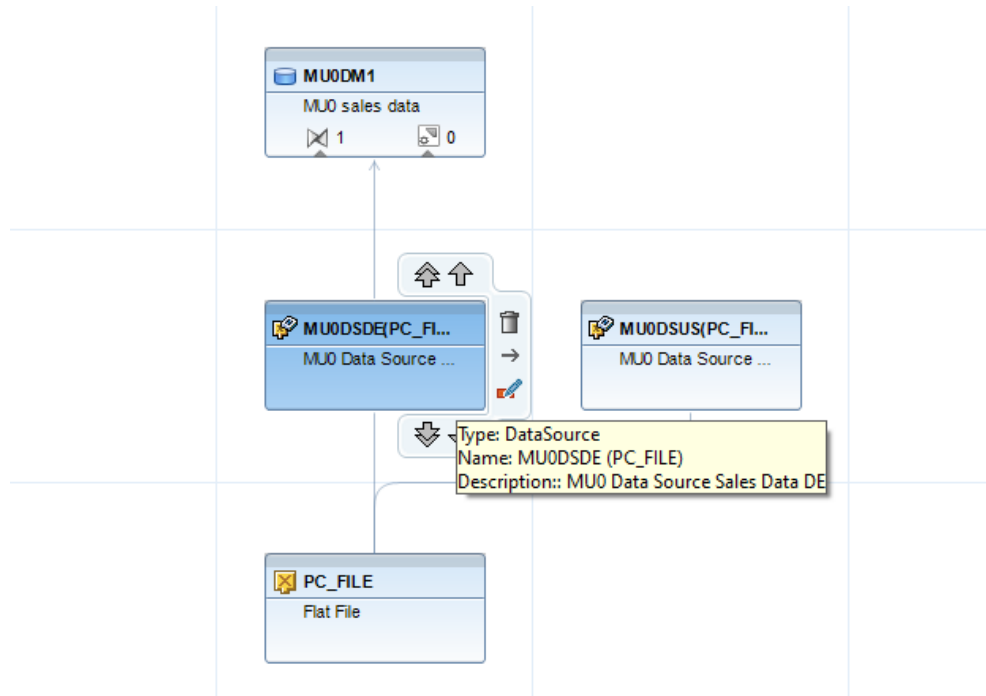


Figure 14

5. **Save** the DataFlow.
6. Create a transformation via double-click on the white transformation symbol.

The 'New Transformation' dialog box is shown with the following fields:

- BW Project:** UHU_900_kale_en
- Target**
 - Object Type:** DataStore Object (advanced)
 - Object Name:** MU0DM1
- Source**
 - Object Type:** DataSource
 - Object Name:** MU0DSDE
 - Source System:** FILE
 - Copy Rules from:** (empty)

Buttons: 'Finish' and 'Cancel'.

7. **Finish.**
8. Click on **Rules** Tab.
9. In the transformation editor, define the transformation rules by mapping the source and target fields. As compared to the transformation of the material master data, we

have in this case not only direct assignment of field/InfoObjects, but in addition more complex rules.

10. First, assign the fields of the DataSource with the fields/InfoObjects in the advanced DataStore Object that can be *directly assigned*.
 - a. **CUSTOMER** to **MU0CUST** (description: [Customer](#))
 - b. **PRODUCT** to **XXXMA###** (description: [XXX### Material](#))
 - c. **REVENUE** to **XXXREV###** (description: [XXX### Revenue](#))
 - d. **SALESQUANTITY** to **XXXQTY###** (description: [XXX### Sales Quantity](#))
 - e. **DISCOUNT** to **MU0DISCNT** (description: [Discount](#))
 - f. **COSTSINUSD** to **MU0COGM** (description: [Cost of Goods Manufactured USD](#))

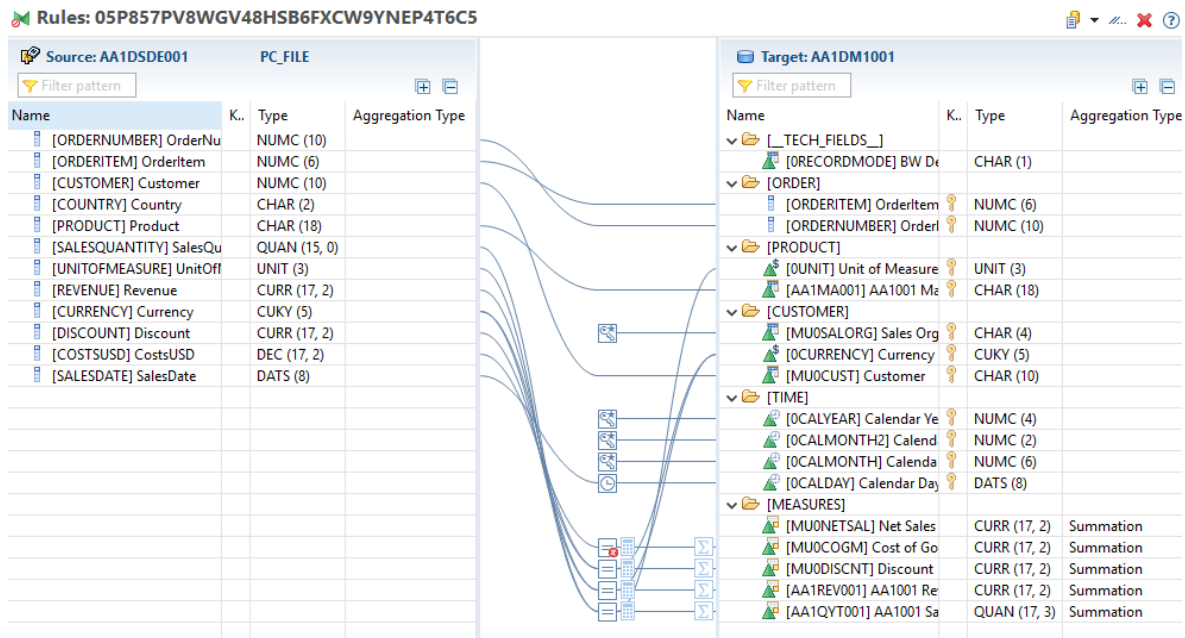


Figure 15

11. The key figure [Cost of Goods Manufactured USD](#) has fixed currency USD. Double-click the = sign for Cost of Goods Manufactured.
12. Since our source field is in USD as well we have to choose **Direct Assignment** in the rule.
13. **Uncheck** Conversion Type

Transformation RSDS AA1DSDE001 PC_FILE -> ADSO AA1DM1001

General

Rules

Technical

Target Fields:

Target Name	Type
[MU0COGM] Cost of Goods M...	CURR (17, 2)

Aggregation Type: Summation

Source Amount Field: COSTSUSD Browse...

Fixed Target Currency: USD

☐ Conversion Type: Browse...

Rule Description:

Type: Direct Assignment

Source Fields:

Source Name	Type
[COSTSUSD] CostsUSD	DEC (17, 2)

Add Field... Remove

Figure 16

14. Save.

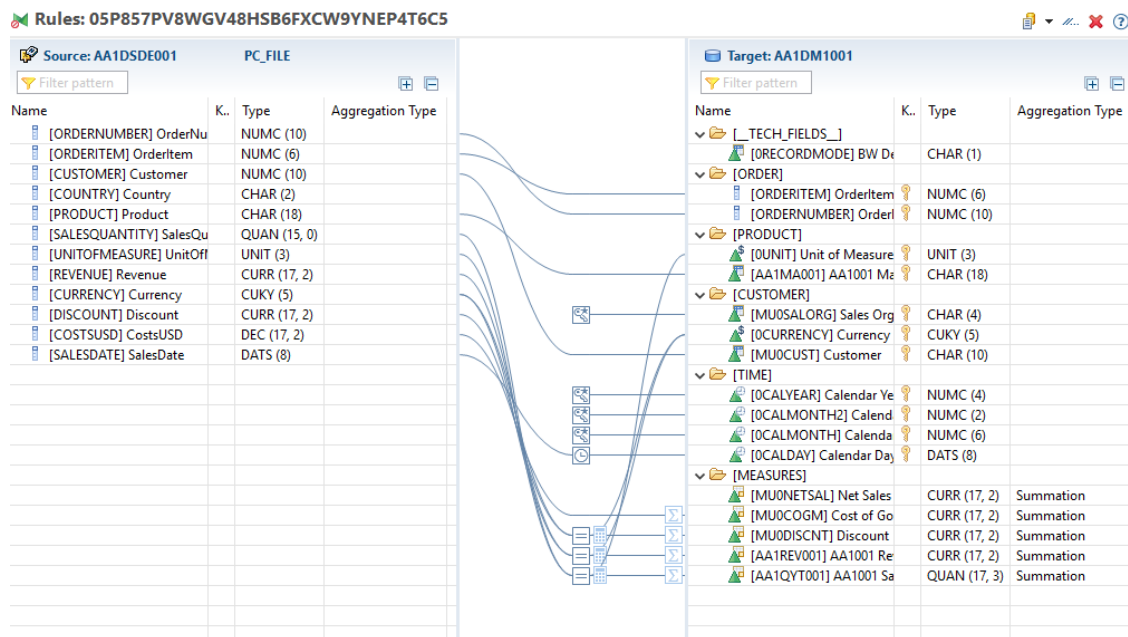


Figure 17

15. The system can derive all *time characteristics* automatically from **SalesDate** as source field. This contains **oCALYEAR** (Calendar Year), **oCALMONTH2** (Calendar month), **oCALMONTH** (Calendar Year/Month), **oCALDAY** (Calendar Day).

16. **Connect** all of the source and target objects as shown below.

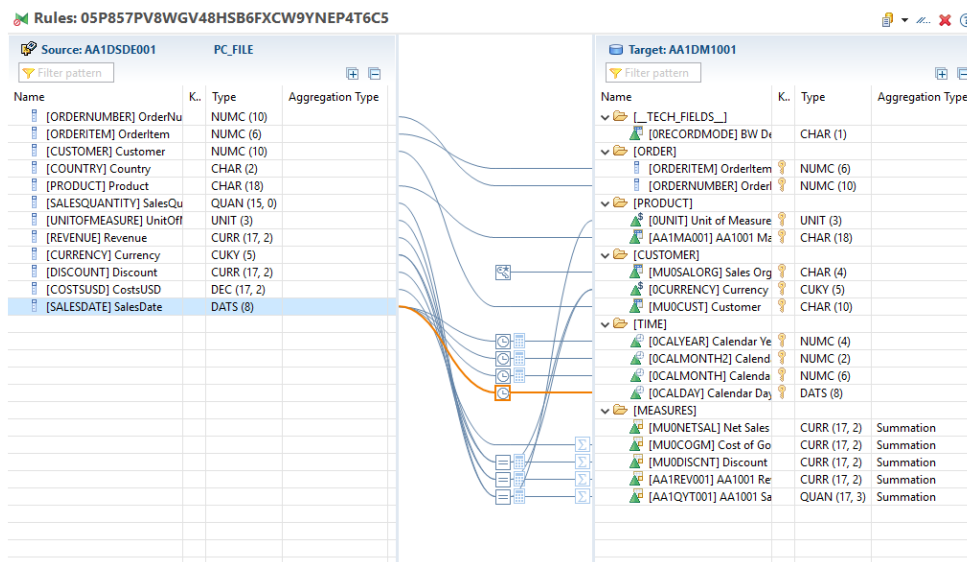


Figure 18

17. Configure the transformation of **Calendar Day** as follows:

Transformation RSDS MU01DSDE001 PC_FILE -> AD50 MU0DSO001

Target Name	Type
[OCALDAY] Calendar Day	DATS (8)

Rule Description:

Type: Time

Source Fields:

Source Name	Type
[SALESDATE] Salesdate	DATS (8)

Use Source Field as: [OCALDAY]

Type of Time Assignment: Direct Assignment

Run Master Data Check for InfoObject: [OCALDAY]

Mapped Source Field: [SALESDATE] Browse...

Figure 19

18. **Save.**

19. Configure **Net Sales** which is defined as a rule of type “**Formula**”:

Net Sales = Revenue - Discount

20. To define this formula, add **REVENUE** and **DISCOUNT** as source fields for the rule (MUoNETSA, Net Sales), then click on the target field **Net Sales** and choose then rule type **Formula**. The net sales is calculated during load process, record by record. Set the options as shown below.

Figure 20

21. **Save.**

22. There is only one InfoObject that does not have a transformation yet.

23. **Sales Organization** is not part of transaction data but in the master data table of the InfoObject Customer (Customer has an attribute Sales Organization).

24. So, the idea is to do a “*master data lookup*” when loading the data to the data mart layer.

25. **Configure Sales Organization.** It is a time-dependent attribute of customer. Therefore, the system must know the date when reading the attribute.

26. Change the **Rule Type** to **Lookup**.

27. Add the **Customer** first.

28. The lookup object is **Customer**.

29. Set the following options.

Figure 21

30. Then the **Salesdate** field will be added automatically in the next step.

31. Click on **Map Fields for Lookup Object**. Fill all of the fields as shown below.

Map Fields of Lookup Object

Lookup Object ▲ MU0CUST

First Step:
Every key field of the lookup object has to be mapped to a field in the transformation source

Select field(s) in the transformation source

Key Field (Lookup Object)	Field (Transformation Source)
▲ [MU0CUST] Customer	[CUSTOMER] Customer
▲ [0CALDAY] Calendar Day	[SALESDATE] Salesdate

Second Step:
Every target field of the rule has to be mapped to a field in the lookup object

Select field(s) in the lookup object

Field (Target)	Field (Lookup Object)
▲ [MU0SALORG] Sales Organizati...	[MU0SALORG] Sales Organiza...

Define the key date for time-dependent master data read

Key Date: from Source Field ▼

Source: SALESDATE Browse... Use Source Field as: 0CALDAY ▼

☐ First day in selected time period
☒ Last day in selected time period

Figure 22

32. Click OK.

33. Your rule should look like this.

Target Fields:

Target Name	Type
▲ [MU0SALORG] Sales Organizati...	CHAR (4)

Rule Description:

Type: Lookup ▼

Source Fields:

Source Name	Type
[CUSTOMER] Customer	NUMC (10)
[SALESDATE] Salesdate	DATS (8)

Add Field... Remove

☒ Perform Conversion Exit
[Lookup Object](#)

☒ Run Master Data Check for InfoObject
[Map Fields of Lookup Object...](#)

ALPHA

▲ MU0CUST Browse...

Figure 23

34. **Save** and **Activate** your transformation (warning can be ignored).

35. **Close** the transformation.

36. Go back to your data flow and **save** and **activate** the data flow.

37. The transformation icon should be green (bowtie).

38. Connect the second **DataSource for US** with the **DataStore Object** - Right-click the **Sales Data ADSO** → **Transformation** → **Create Transformation**.

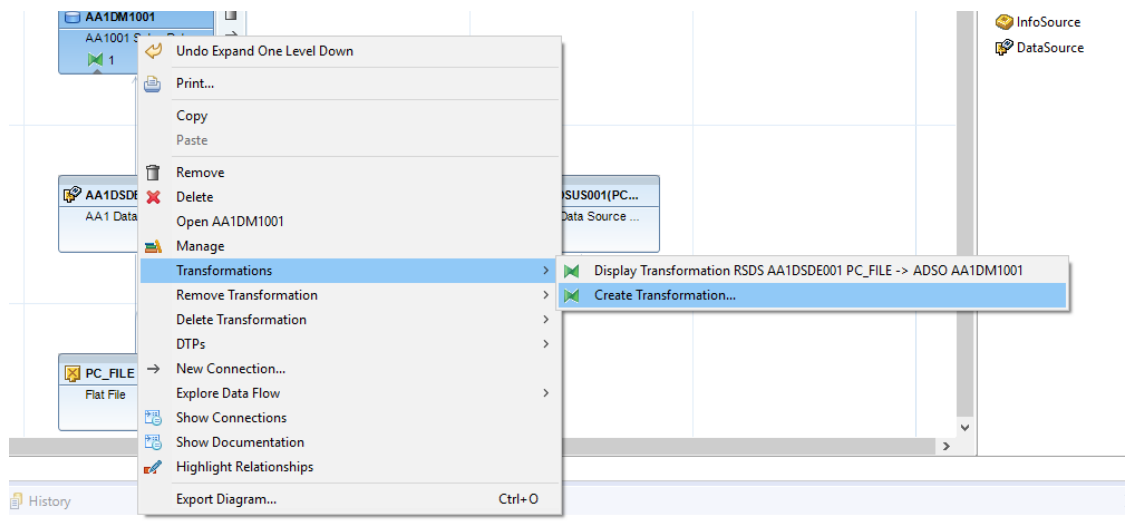


Figure 24

39. On the following popup, name of the DataSource in **XXXDSUS###**.

40. **Copy Rules** from your **XXXDSDE###** to **XXXDM1###** transformation.

41. **Finish**.

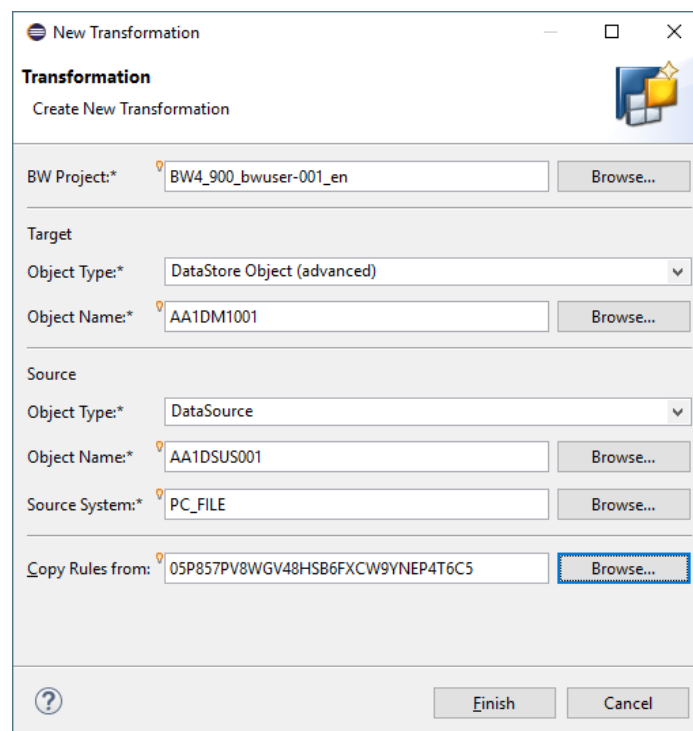


Figure 25

42. **Save** and **Activate** your transformation (warning can be ignored).
43. **Close** transformation.

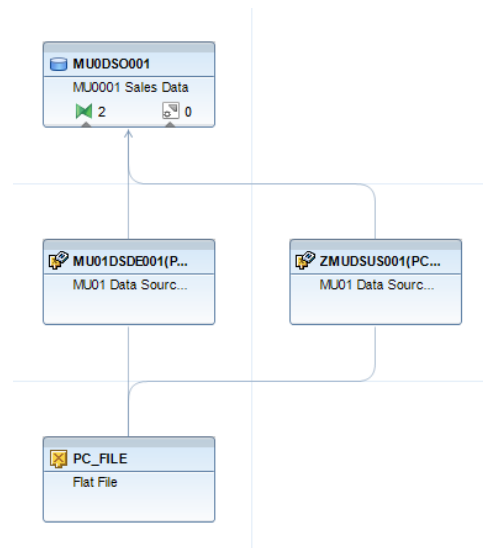


Figure 26

44. **Save** and **activate** the data flow.

- Question 4:** Why did we choose Direct Assignment for Customer?
- Question 5:** Can one source object be mapped to multiple target objects? Provide an example.
- Question 6:** Why does the Cost of Good Manufactured not need a currency key?
- Question 7:** What will be the final currency of Net Sales?
- Question 8:** Explain, in detail, the transformation rule for Sales Organization.
- Question 9:** Why does the Sales Data aDSO need two transformations?
- Question 10:** Why is the ordernumber a field and not an InfoObject?

5. Create a Data Transfer Process and load data.

Jan creates a *Data Transfer* Process between the data acquisition layer (sales transaction Data Source from Global Bike Germany and US) and the data mart layer (aDSO). Then, he loads the data into the data warehouse.

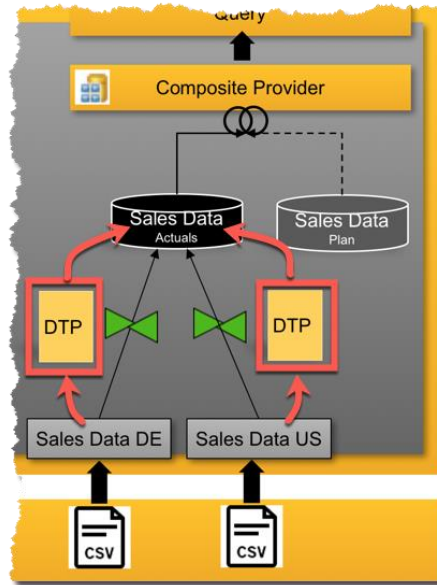


Figure 27

1. Open **Eclipse**.
2. Navigate to your InfoArea folder **XX** → **XXX** → **XXX###** (name: **Student ###**). Open your Data Flow from previous step with technical name **XXXDFSD###**.
3. Create a Data Transfer Process (DTP) between the left DataSource (**Sales Data DE ###**) and the aDataStore Object (**XXX### Sales Data**) (**draw a line** using the arrow).

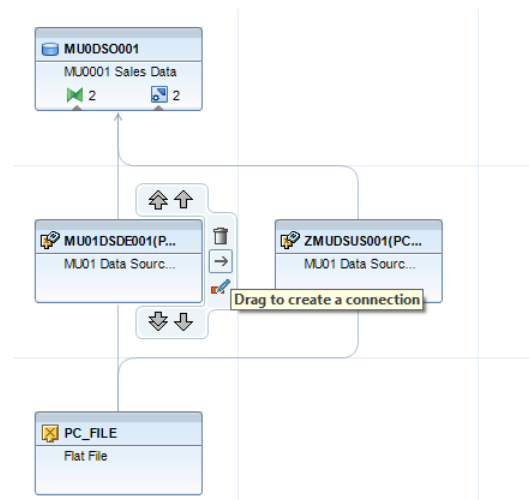


Figure 28

4. Double-click on **DTP**.
5. **Next** → **Finish**.
6. Configure the following settings:

- a. Extraction Mode: **Full**
- b. **Extraction** tab
 - (1) File Name: Browse for your *Sales data DE.csv* file.

Figure 29

7. **Save** and **Activate** the DTP.
8. Click on Start Execution of DTP (top right).
9. Click on  **Execute**.
10. **Enter** → **Allow** → **Yes**.

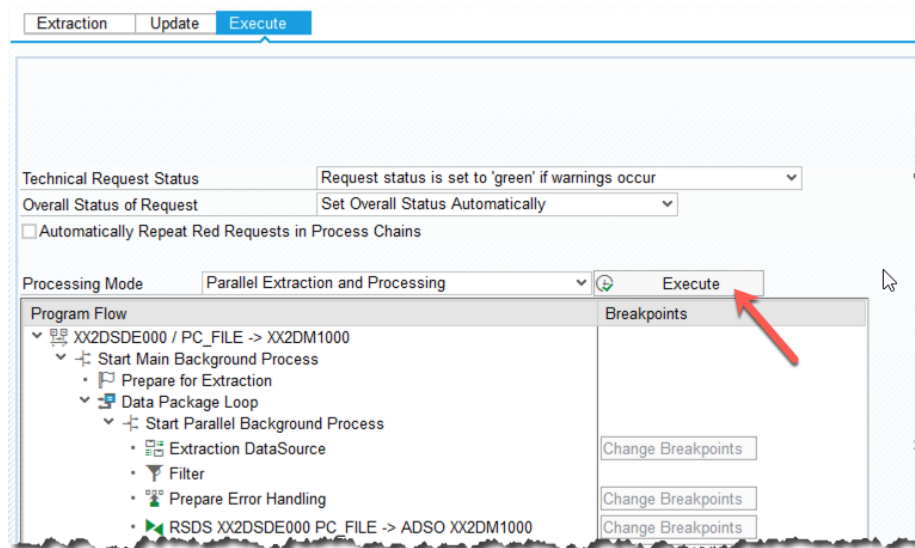


Figure 30

11. Check the monitor if everything was ok (refresh may be necessary .
12. Click through any popups. Be sure that 86,816 transactions have been loaded.

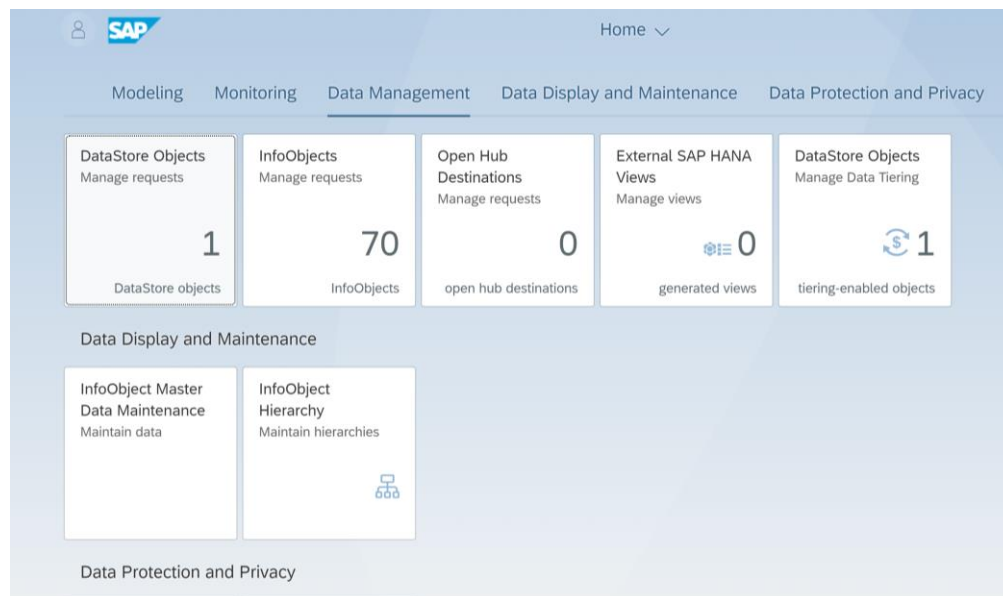


Figure 32

22. On the following screen, you see an overview over all requests loaded into your aDSO. The requests are grouped by day when you executed the DTP.

Standard * ▼					
Not Filtered ▼					
DataStore Objects (1) ⚙️ 📄 ▼					
DSO (advanced)	Aggregated Status	InfoArea	Successful Requests	Running Requests	Failed Requests
MU0 Sales Data aDSO MU0DM1	OK (ok)	Master Folder for BW/4HANA 2.0 Curriculum DO NOT EDIT (MU0)	2	0	0 >

Figure 33

23. Click your aDSO

24. Select the **Request** and then **Activate**.

MU0 Sales Data aDSO MU0DM1						Refresh	Further Actions	Open Request	
☒ Default View ☐ Advanced View									
Requests (2 of 2)						☐ ☒ ☐ ☐ Select Status			
Request	Storage	Records	Last Action	Log	Last Action On				
☐ {2021-05-21 01:36:16 000000 CST}	Inbound Table	84,194	☒ Loading	Log	21.05.2021, 12:07:19				
☐ {2021-05-21 01:10:56 000000 CST}	Inbound Table	86,816	☒ Loading	Log	21.05.2021, 11:42:20				

Figure 34

Details of Load Request {2021-05-21 01:36:16 000000 CST}

Refresh

DTP Monitor

Display Request Data

Request Information

DataStore Object: MU0DM1

DataStore Object Description: MU0 Sales Data aDSO

ID: {2021-05-21 01:36:16 000000 CST}

Status: Overall and technically OK

Storage: Inbound Table

Records: 84,194

DataSource: MU0DSUS FILE

Description: MU0 DataSource Sales Data US

DTP Properties

Name: DTP_0002TLMGJRD4TYIKSJ2RH46DJ

Description: RSDS MU0DSUS FILE -> ADSO MU0DM1

Extraction Mode: Full

Processing Mode: Serial SAP HANA Execution in Dialog Process

Package Size: 1,000,000

Request Aggregation: SUM

Request Start Time: 21.05.2021, 12:06:14

Request Finish Time: 21.05.2021, 12:07:21

Request Runtime: 1m 6s

Selection:

Activate

Delete

Figure 35

25. After successful activation, the request overview looks like this. You may need to Refresh.

MU0 Sales Data aDSO MU0DM1						Refresh	Further Actions	Open Request	
☒ Default View ☐ Advanced View									
Requests (1 of 1)						☐ ☒ ☐ ☐ Select Status			
Request	Storage	Records	Last Action	Log	Last Action On				
☐ {2021-05-21 01:48:26 000006 CST}	Active Data Table	171,010	☒ Activating	Log	21.05.2021, 12:18:31				

Figure 36

26. Your data is now loaded and can be used for analytics from the aDSO. Go back to your Data Flow (in Eclipse). **Save** and **activate** it.

27. **Close** the Data Flow Object. In the ETL process what does DTP represent?

28. Open your aDSO in Eclipse. Click on Data Preview. You should confirm that the total Cost of Goods Manufactured is \$953,354,225.28

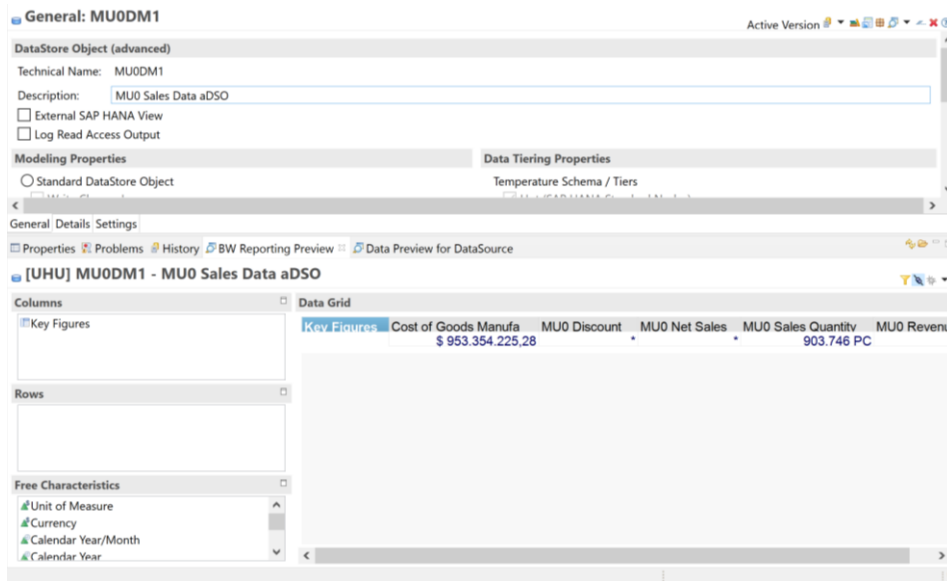


Figure 37

Question 11: Why did you have to execute two DTPs?

Question 12: What would happen if you executed a DTP twice by mistake?

Question 13: What is the source object of the DTP?

Question 14: What is the target object of the DTP?

Question 15: In these DTP, what type of data are we loading (in the context of the snowflake schema)?

6. Create a CompositeProvider.

Jan uses Eclipse to create the virtualization layer of the data warehouse with a *CompositeProvider*.

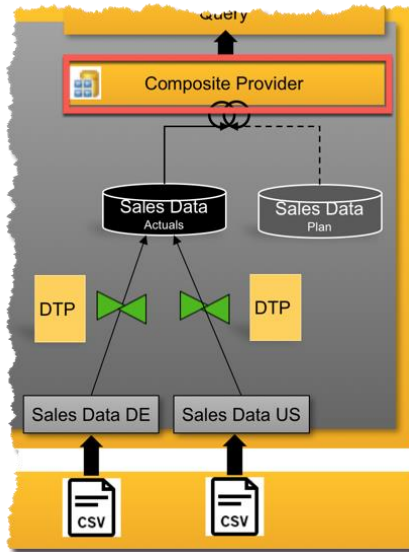


Figure 38

1. Start **Eclipse** and open your BW project: **Project Explorer** → **your BW-System** → **BW Repository**.
2. Open your BW project in Eclipse and navigate to the InfoArea assigned by your instructor, which has the technical name **XXX###** and description **Student ###**.
3. Choose **New** → **CompositeProvider**.

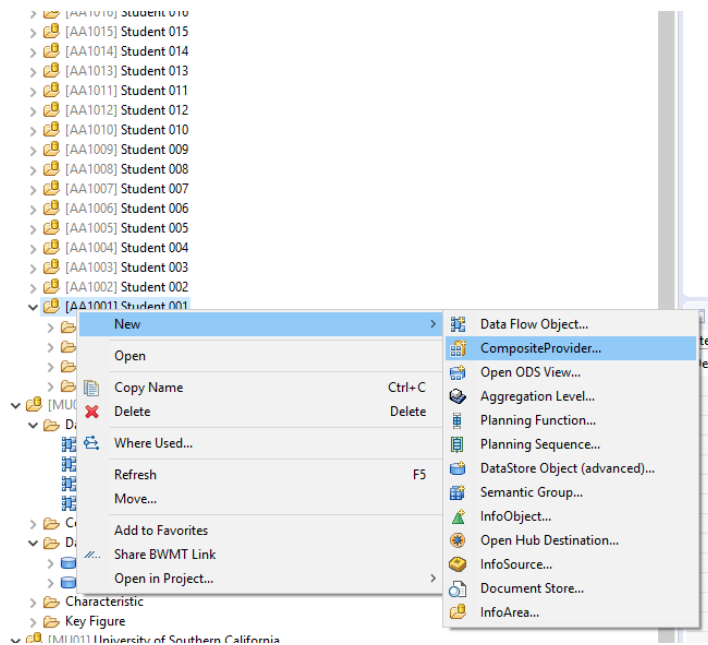
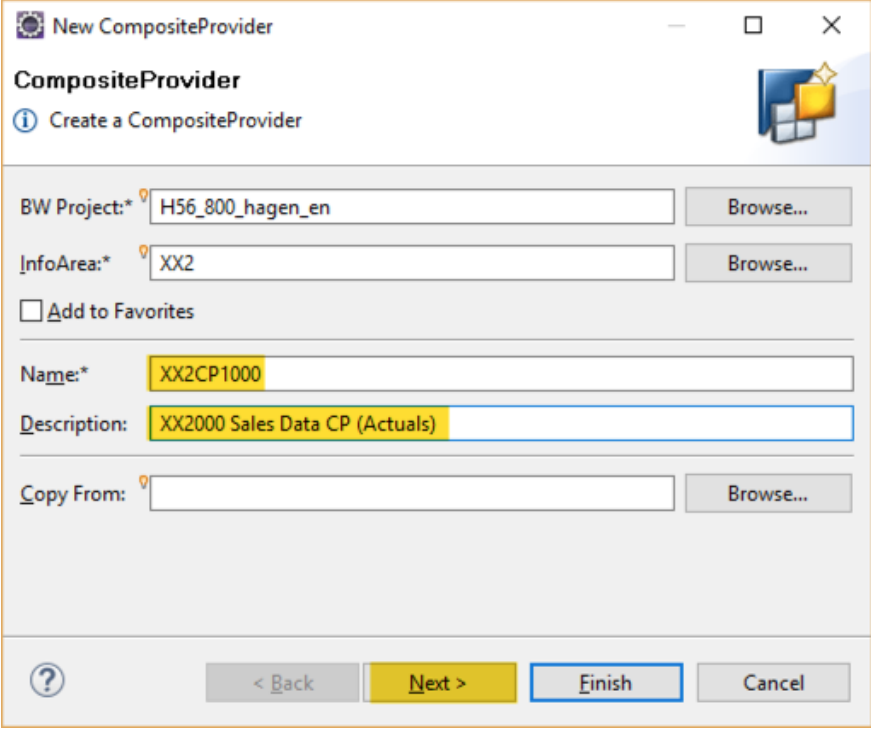


Figure 39

- On the following screen, choose as name **XXXCP1###** and as description **XXX### Sales Data Composite Provider**.



The screenshot shows a 'New CompositeProvider' dialog box. The title bar says 'New CompositeProvider'. Below the title bar, there's a section 'CompositeProvider' with a sub-label 'Create a CompositeProvider'. The main area contains several input fields: 'BW Project:*' with the value 'H56_800_hagen_en', 'InfoArea:*' with the value 'XX2', 'Name:*' with the value 'XX2CP1000', and 'Description:' with the value 'XX2000 Sales Data CP (Actuals)'. There are 'Browse...' buttons next to the first three fields. Below these fields is a checkbox 'Add to Favorites' which is unchecked. At the bottom, there's a 'Copy From:' field with a 'Browse...' button. The bottom of the dialog has a navigation bar with a question mark icon, '< Back', 'Next >' (highlighted in yellow), 'Finish', and 'Cancel'.

Figure 40

- Choose **Next** to add your aDataStore Object (technical name: **XXXDM1###**) as part provider in the **Union** section.

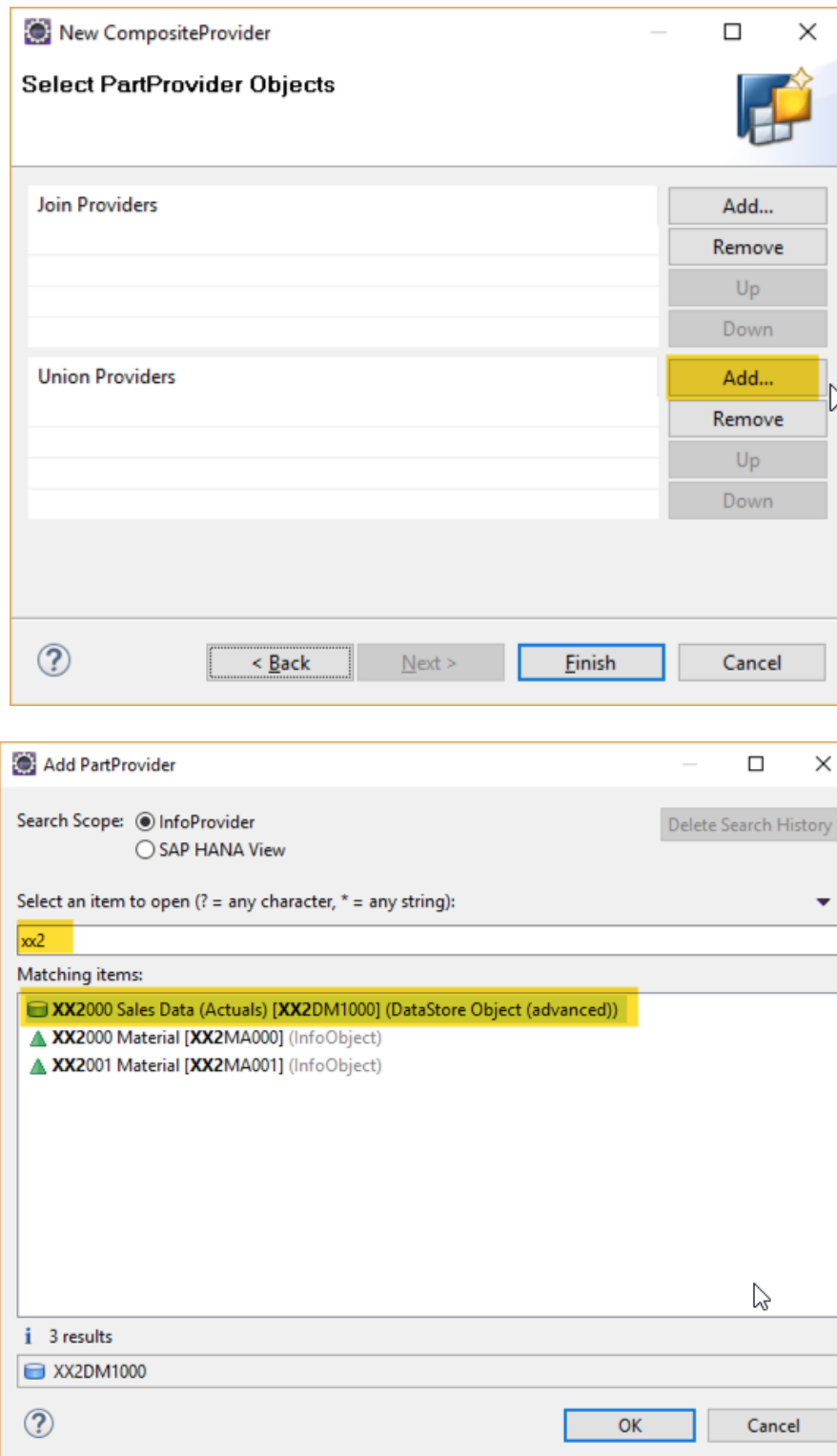


Figure 41

6. Choose **Finish** to create the CompositeProvider (CP) and open the editor for this object. Save.

7. Extend the CompositeProvider

Jan uses Eclipse to define *join* and *union conditions* to link the part providers that make the CompositeProvider.

1. Start **Eclipse** and open your BW project: **Project Explorer** → **your BW-System** → **BW Repository**.
2. Open your CompositeProvider from previous step with technical name **XXXCP1###**.
3. On the tab **Scenario**, you define in general the join and union conditions to link the part providers that make the CompositeProvider.

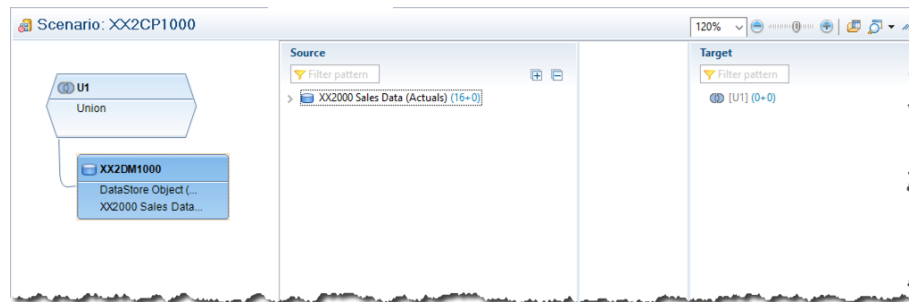


Figure 42

4. In our case, we have only one InfoProvider.
5. Just **drag** from source to target to assign the fields to the target structure (a block, not field by field).

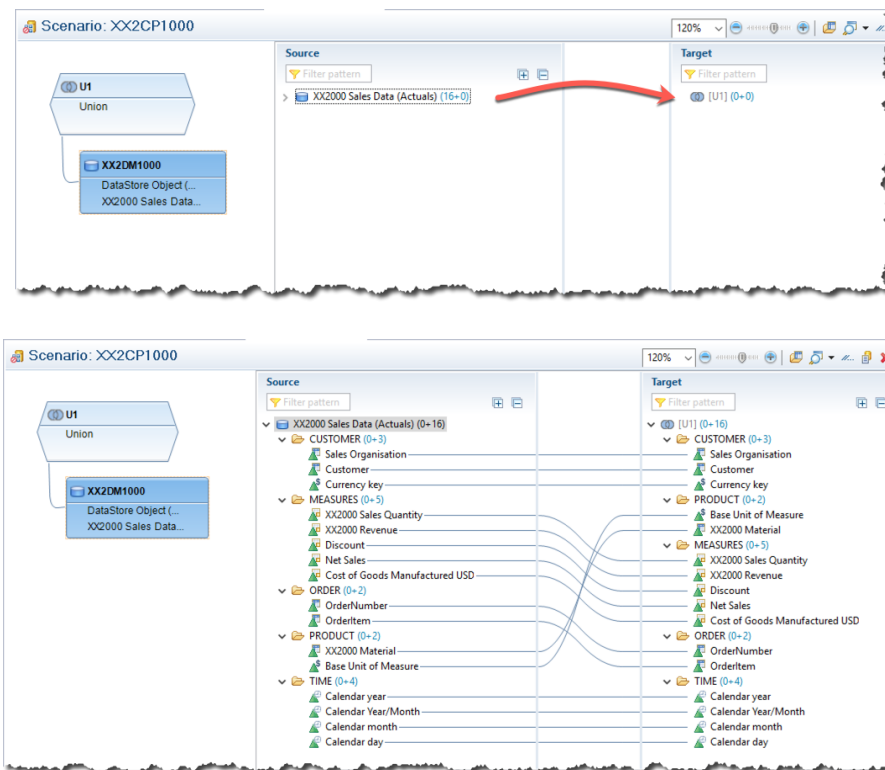


Figure 43

6. **Save and activate** your CompositeProvider.

Question 16: What is the difference between an aDSO and a composite provider?

Question 17: Is the data in a composite provider persistent?

8. Change properties of the output structure of the CompositeProvider.

Jan uses Eclipse to extend the CompositeProvider. For this, he changes some *properties* of the *output* structure of the CompositeProvider.

1. Start **Eclipse** and open your BW project: **Project Explorer** → **your BW-System** → **BW Repository**.
2. Open your CompositeProvider from previous step with technical name **XXXCP1###**.
3. On tab **Output** we can change some properties of the output structure of the CP, e.g. names and we can enrich the CP with navigation attributes of the underlying InfoObjects.

4. First, change the description of InfoObject **XXXMA###** to **Product**, as this is the term the sales department would prefer to use.

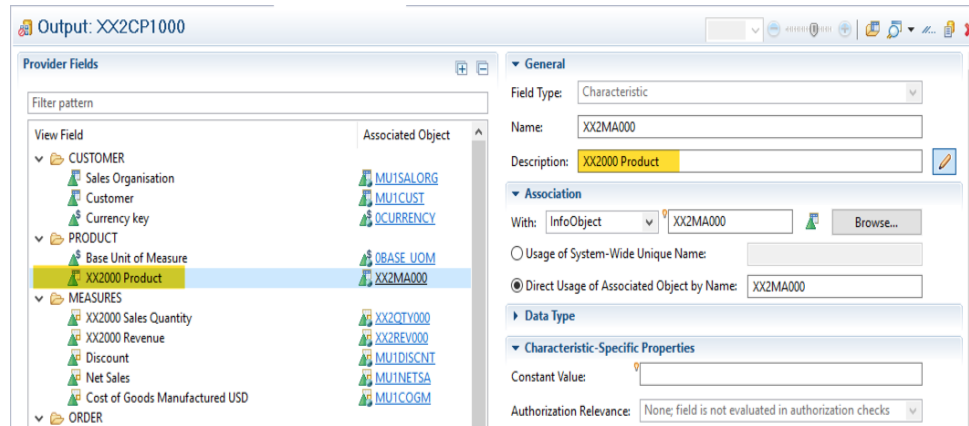


Figure 44

5. In the context menu (right-click) of InfoObject Product (XXXMA###) choose **Navigation Attributes**.
6. Switch on the navigation attributes Product Category (**XXXPC###**) and Division (**MUoDIVN**).

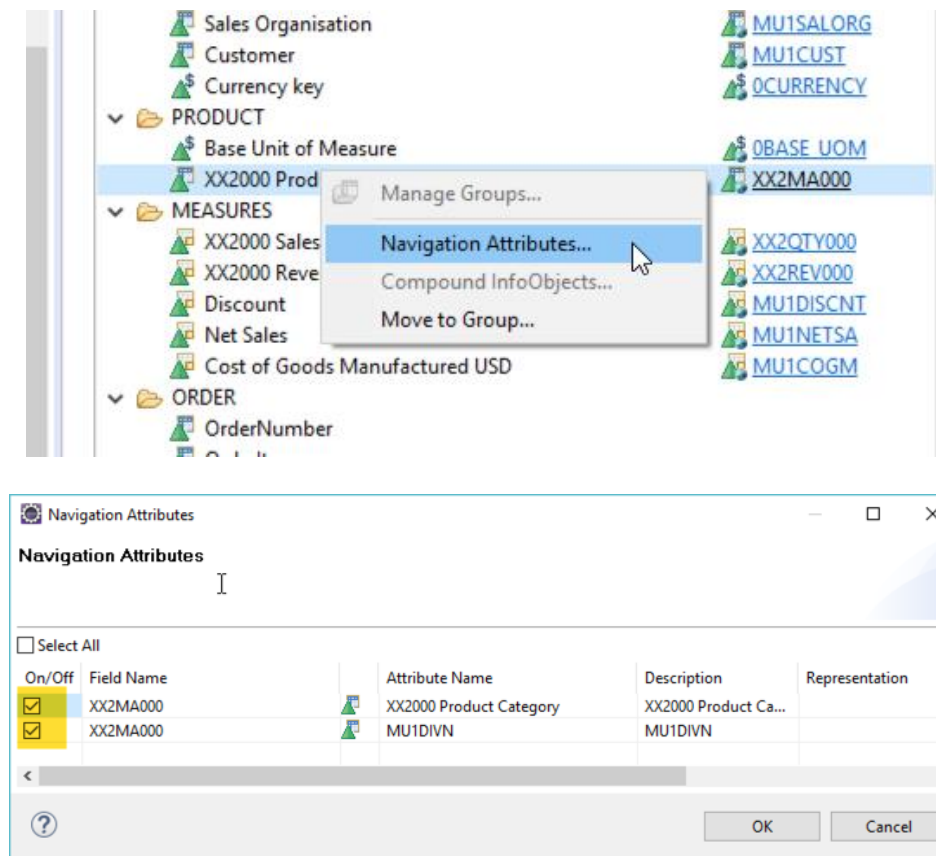


Figure 45

7. In the context menu of InfoObject Customer ([MUoCUST](#)) choose **Navigation Attributes**.
8. Switch on the navigation attribute Country ([MUoCOUNTRY](#)).

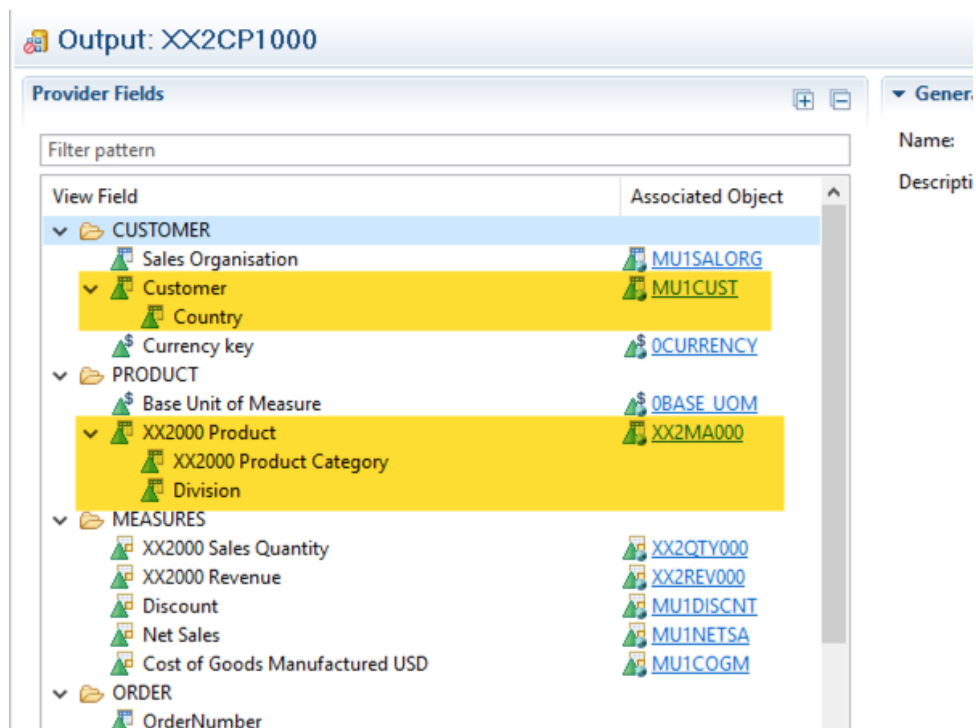
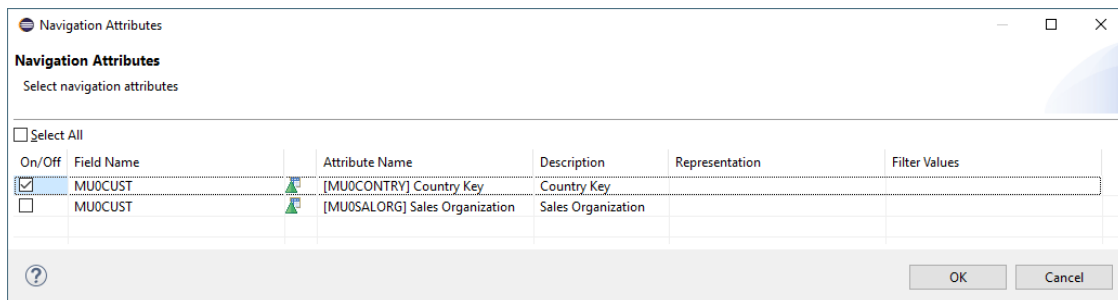


Figure 46

9. Finally, **save** and **activate** your CompositeProvider.
10. In case of success, you get this message:

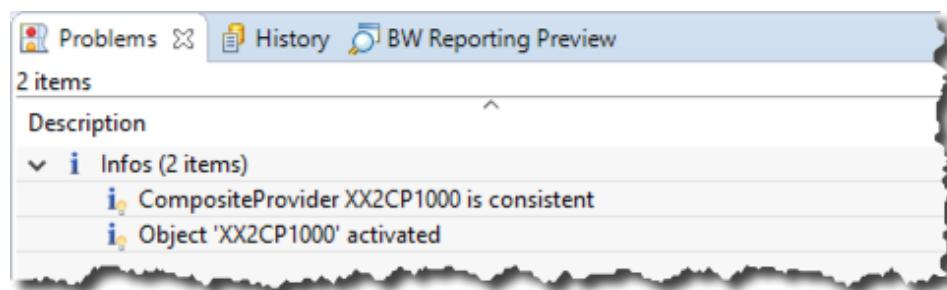
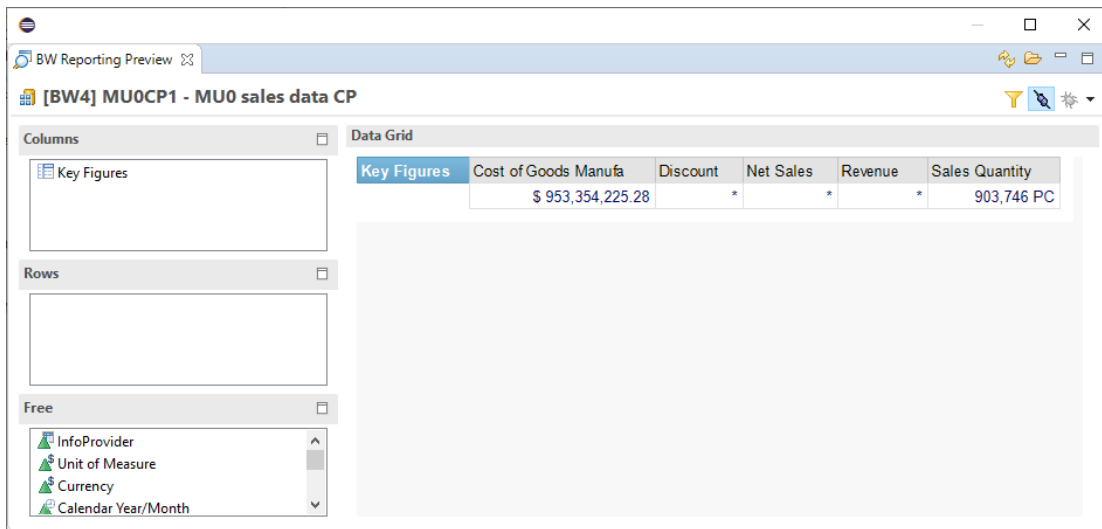


Figure 47

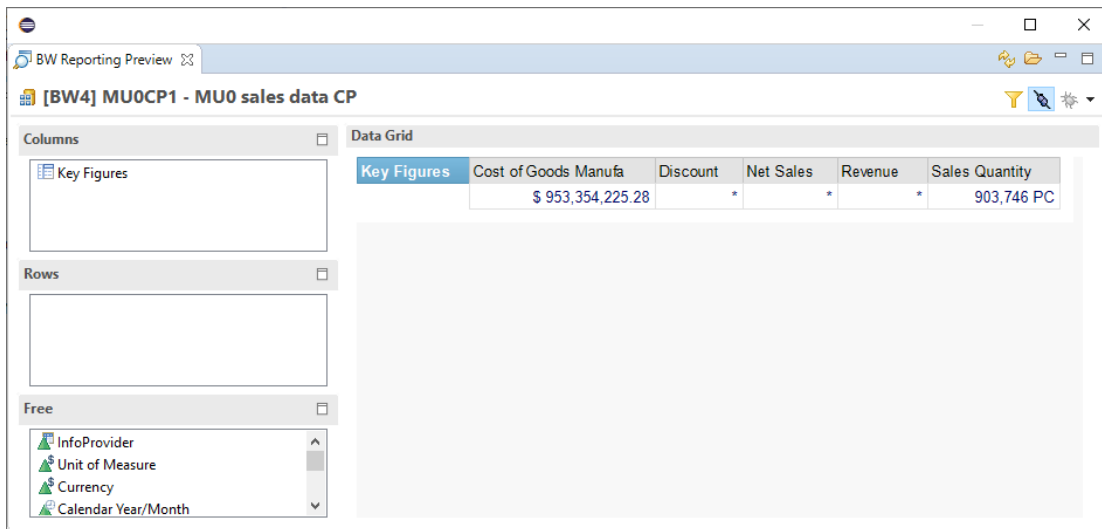
11. You can test your CP with the “Data Preview”. Click on the upper right side on  and choose **Data Preview**.



Key Figures	Cost of Goods Manufa	Discount	Net Sales	Revenue	Sales Quantity
	\$ 953,354,225.28	*	*	*	903,746 PC

Figure 48

12. Without any drill down:



Key Figures	Cost of Goods Manufa	Discount	Net Sales	Revenue	Sales Quantity
	\$ 953,354,225.28	*	*	*	903,746 PC

Figure 49

13. Drill down by customer: **Drag and drop** **Customer** into **Rows**.

[BW4] MU0CP1 - MU0 sales data CP

Key Figures	Cost of Goods Manufa	Discount	Net Sales	Revenue	Sales Quantity
Customer					
Airport Bikes	\$ 37,152,220.20	1,656,311.81 EUR	53,554,075.94 EUR	55,210,387.75 EUR	35,049 PC
Alster Cycling	\$ 35,489,583.65	1,598,084.22 EUR	51,671,378.18 EUR	53,269,462.40 EUR	33,931 PC
Bavaria Bikes	\$ 96,447,372.41	7,257,248.95 EUR	137,887,600.20 EUR	145,144,849.15 EUR	85,864 PC
Beantown Bikes	\$ 64,070,440.86	\$ 5,480,569.91	\$ 104,130,788.35	\$ 109,611,358.26	63,012 PC
Big Apple Bikes	\$ 39,842,321.21	\$ 2,046,239.36	\$ 66,161,728.77	\$ 68,207,968.13	38,250 PC
Capital Bikes	\$ 53,853,966.12	2,428,795.29 EUR	78,531,022.78 EUR	80,959,818.07 EUR	51,719 PC
Cruiser Bikes	\$ 37,126,315.76	1,666,227.54 EUR	53,874,671.45 EUR	55,540,898.99 EUR	35,108 PC
DC Bikes	\$ 33,352,652.58	\$ 1,709,351.62	\$ 55,269,030.15	\$ 56,978,381.77	32,274 PC
Drahtesel	\$ 30,057,258.50	0.00 EUR	44,656,137.94 EUR	44,656,137.94 EUR	27,588 PC
Fahrrpott	\$ 30,702,209.66	0.00 EUR	45,481,283.36 EUR	45,481,283.36 EUR	29,707 PC
Furniture City Bikes	\$ 17,419,391.43	\$ 0.00	\$ 29,802,418.14	\$ 29,802,418.14	16,804 PC
Motown Bikes	\$ 35,279,545.74	\$ 1,810,349.91	\$ 58,534,649.64	\$ 60,344,999.55	33,792 PC
Neckarad	\$ 43,644,036.95	1,973,240.88 EUR	63,801,413.72 EUR	65,774,654.60 EUR	40,133 PC

Figure 50

14. You have now modeled, designed, implemented, and populated SAP BW/4HANA 2.0 Data Warehouse! It is now ready for analytics!

Question 18: What object in all of SAP BW/4HANA represents the snowflake schema?

Question 19: List the process (at least 10 steps) of building and populating a BW/4HANA 2.0 data warehouse. Refer to all previous exercises in Chapters 3 and 4.

Question 20: Now that you have modeled, built and populated a data warehouse in SAP BW/4HANA 2.0, list one advantage to snowflake schemas that we had discussed in class but did not implement in our data warehouse.